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ENGINEERING



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CSA1714 ARTIFICIAL INTELLIGENCE

Edge Intelligence: AI Without the Cloud

CO5 ASSESSMENT TOOL 2 TECHNICAL PRESENTATION

• Unit V — Agents, Case Studies & Real-Time Projects

LOW LATENCY • PRIVACY • REAL-TIME AI

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Presentation Objectives



1

01 • Understand

Define Edge AI and distinguish it from cloud-based AI.

2

02 • Architecture

Explain the end-to-end edge intelligence pipeline.

3

03 • Techniques

Introduce inference optimization, quantization and model deployment.

4

04 • Applications

Connect Edge AI with real-time IoT, healthcare, smart cities and industry.

5

05 • Engineering View

Evaluate latency, privacy, energy, accuracy and security trade-offs.



What Is Edge Intelligence?

AI inference and, in some systems, learning performed close to where data is generated.

Core Idea

Move useful computation from a distant centralized cloud toward devices, gateways and nearby edge servers. This reduces dependence on continuous cloud connectivity.

Why It Matters

Edge systems are designed around real-time response, constrained compute, energy efficiency, privacy and resilient operation.

**Sensors /
Cameras**

Edge Device

AI Inference

Action

Cloud AI vs. Edge AI

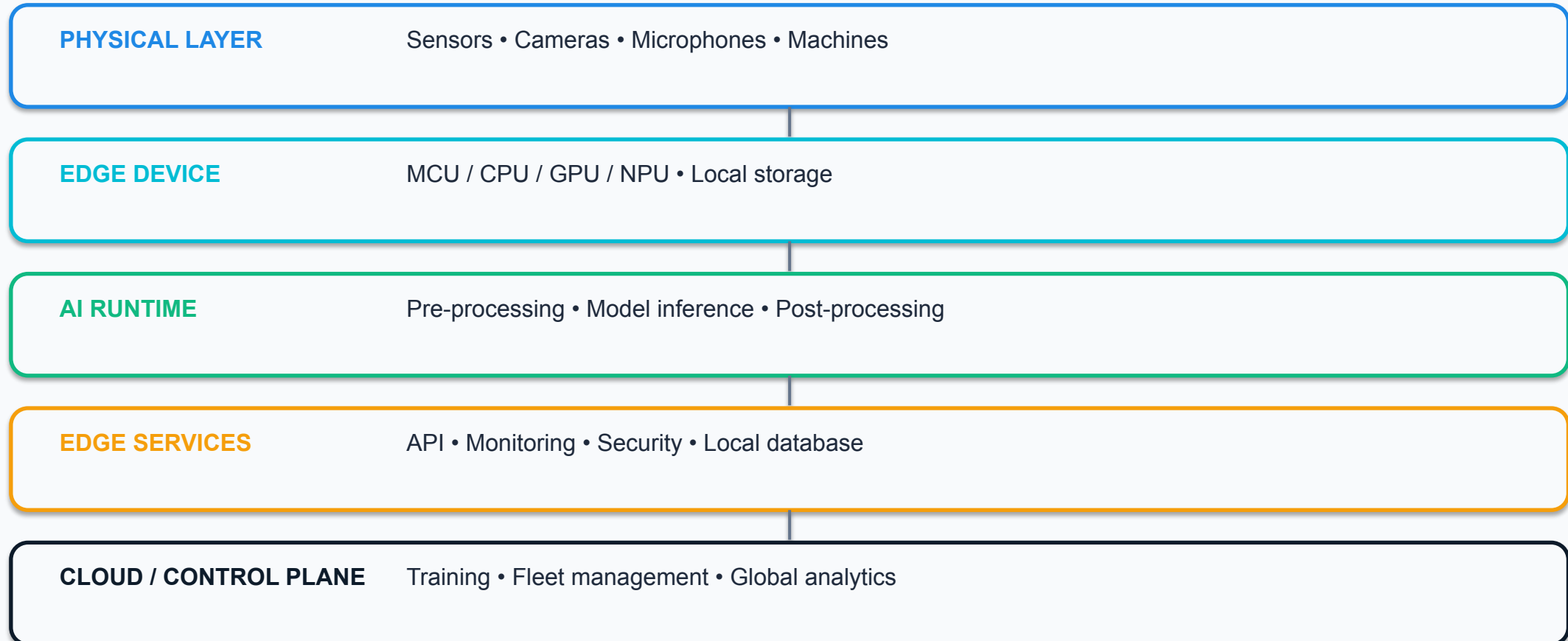
The engineering choice is not 'cloud or edge'—many practical systems use both.



Dimension	Cloud-Centric AI	Edge AI
Latency	Network round-trip may add delay	Local inference can respond faster
Bandwidth	Raw/processed data sent upstream	Less data needs to leave the device
Privacy	Data often leaves local environment	Sensitive data can remain local
Compute	Large centralized resources	Limited CPU/GPU/NPU resources
Availability	Depends on network/service	Can continue during weak connectivity
Energy	Device communication + cloud compute	Local compute must be energy-aware

Reference Architecture

A practical Edge AI stack from physical data to intelligent action.

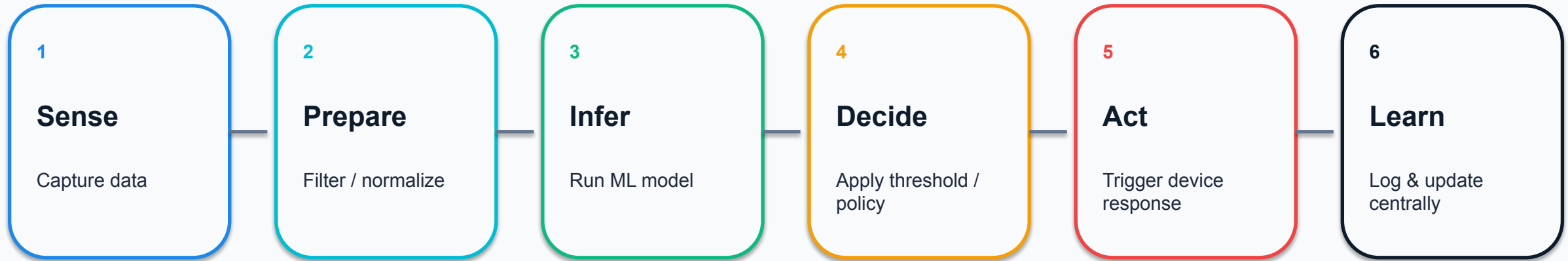


Key principle: send models/data to the edge only when the performance, privacy or reliability benefit justifies the resource cost.



How an Edge AI System Works

A real-time inference loop.



Example

A smart camera detects an unsafe event locally, decides whether confidence is high enough, then triggers an alarm—without uploading the full video stream.



Making AI Small Enough for the Edge

Model optimization balances accuracy, latency, memory and energy.

Quantization

Represent weights/activations with lower precision such as INT8 to reduce memory and compute.

Pruning

Remove low-value parameters or connections to reduce model size and operations.

Knowledge Distillation

Train a smaller student model to reproduce useful behavior from a larger teacher model.

Hardware Acceleration

Use GPU/NPU/TPU/AI accelerators to execute supported operations efficiently.

Early Exit / Cascades

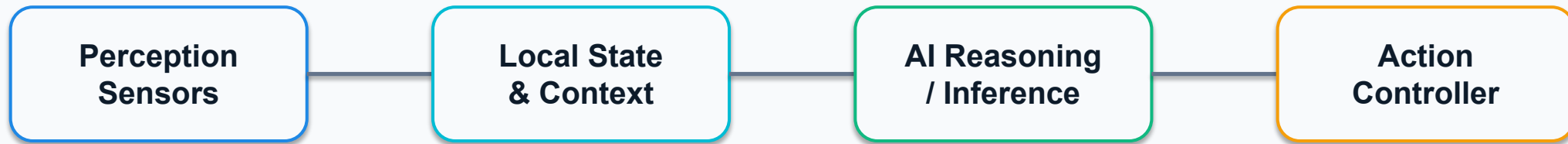
Use a cheap model first; invoke a heavier model only for uncertain cases.

Compiler Optimization

Convert models into hardware-friendly runtimes and optimized kernels.

Edge AI as an Intelligent Agent

Connection to Unit V: perception → reasoning → action under resource constraints.



Autonomy

Makes local decisions without a cloud round trip.

Adaptation

Uses current context and sensor feedback to select actions.

Coordination

Can exchange summaries/models with nearby or cloud agents.

Real-World Applications



Smart Healthcare

Wearables, patient monitoring, privacy-aware analytics

Smart Cities

Traffic cameras, pedestrian safety, environmental sensing

Industry 4.0

Predictive maintenance, quality inspection, robotics

Autonomous Systems

Drones, vehicles, navigation and obstacle detection

Retail & Security

On-device vision, anomaly detection, access control

Agriculture

Crop monitoring, disease detection, irrigation decisions



Mini Case Study: Smart Camera at the Edge

Illustrative engineering design for real-time safety monitoring.



Design Goal

Minimize response time while keeping video local whenever possible.

Metrics

Latency • FPS • accuracy • RAM • power • false alarms

Engineering Trade-off

A smaller model may be faster but can reduce detection accuracy.

Security, Privacy & Reliability

Moving AI to the edge reduces some risks—but introduces new attack surfaces.



Privacy

Raw sensor data can remain local; only events, embeddings or summaries may be shared.

Model Security

Protect model files, keys and update channels from tampering and extraction.

Device Security

Secure boot, hardware roots of trust and access control protect the edge node.

Adversarial Inputs

Attackers may manipulate sensor inputs or exploit model weaknesses.

Fleet Management

Thousands of devices need safe model updates, monitoring and rollback.

Resilience

Local operation can improve availability during intermittent connectivity.

Advantages & Limitations



Advantages

- Lower latency for time-critical decisions
- Reduced bandwidth consumption
- Better data locality and privacy
- More resilient operation with weak connectivity
- Scalable distributed intelligence

Limitations

- Limited CPU/GPU/NPU, memory and battery
- Model updates are harder across large fleets
- Hardware heterogeneity complicates deployment
- Local devices can be physically attacked
- Accuracy–latency–energy trade-offs must be managed

Future Direction

Where Edge Intelligence is heading.



On-device Generative AI

Smaller language/vision models running locally

Federated Learning

Collaborative model improvement without centralizing raw data

6G + Edge AI

Ultra-low-latency connectivity and distributed intelligence

AI Accelerators

Dedicated NPUs and energy-efficient heterogeneous compute

Agentic Edge Systems

Multiple local agents planning, coordinating and acting

Sustainable Edge AI

Optimize accuracy per watt, not accuracy alone



Key Takeaways

- ✓ Edge AI places intelligence close to where data is generated.
- ✓ It improves responsiveness, bandwidth efficiency and data locality.
- ✓ Successful deployment depends on model optimization and hardware awareness.
- ✓ Security, fleet management and energy efficiency are first-class engineering concerns.
- ✓ The future is hybrid: edge devices, edge servers and cloud systems working together.



References – IEEE Papers (2022–2026)

APA-style entries; selected for technical relevance and searchable in Google Scholar.

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Thank You

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